

This article lists verifiable spaceflight-related accidents and incidents resulting in human death or serious injury. These include incidents during flight or training for crewed space missions and testing, assembly, preparation, or flight of crewed and robotic spacecraft. Not included are accidents or incidents associated with intercontinental ballistic missile (ICBM) tests, death or injury to test animals, uncrewed space flights, rocket-powered aircraft projects of World War II, or conspiracy theories about alleged unreported Soviet space accidents.

As of April 2026, 19 people have died during spaceflights that crossed or were intended to cross the boundary of space as defined by the United States, that being 50 mi (80 km) above sea level. Astronauts have also died while training for space missions, such as the Apollo 1 launch pad fire that killed an entire crew of three. There have also been some non-astronaut deaths during spaceflight-related activities.

As of April 2026, more than 188 people have died in spaceflight-related incidents. In addition, astronaut Peter Siebold is the only person in human history to survive an in-flight spacecraft breakup; on 31 October 2014, Siebold survived a 15 kilometer (50,000 ft) fall without supplemental oxygen in only a thin flight suit, unbuckling from his ejection seat and deploying his parachute at 17,000 feet. From the breakup, Siebold suffered serious injuries to his eyes, face, right arm and the right side of his chest, which required multiple surgeries in the weeks following the crash. Despite his injuries, Siebold made a full recovery.

Astronaut fatalities

During spaceflight

As of 17 April 2026, there have been five incidents in which a spacecraft in flight suffered crew fatalities, killing a total of 15 astronauts and 4 cosmonauts. Of these, two had reached the internationally recognized edge of space (100 km or 62mi above sea level) when or before the incident occurred, one had reached the U.S. definition of space at 266,000 ft, and one was

planned to do so. In each of these accidents, the entire crew was killed. As of April 2026, a total of 791 people have flown into space and 19 of them have died in related incidents. This sets the current statistical fatality rate at 2.4 percent.

NASA astronauts who died on duty are memorialized at the Space Mirror Memorial at the Kennedy Space Center Visitor Complex in Merritt Island, Florida. Cosmonauts who died on duty under the Soviet Union were generally honored by burial at the Kremlin Wall Necropolis in Moscow. No Soviet or Russian cosmonauts have died during spaceflight since 1971.

During training or testing

In addition to accidents during spaceflights, 11 astronauts, test pilots, and other personnel have been killed during training or tests.

Non-fatal incidents during spaceflight

Apart from actual disasters, 40 missions resulted in some very near misses and also some training accidents that nearly resulted in deaths.

Non-fatal training accidents

Spaceflight-related accidents and incidents during assembly, testing, and preparation for flight of crewed and uncrewed spacecraft have occasionally resulted in injuries or the loss of craft since the earliest days of space programs. 35 accidents since 2009.

Non-astronaut fatalities

Fatalities caused by rocket explosions

This list excludes deaths caused by military operations, either by deliberate detonations, or accidental during production – for example German V-2 rockets reportedly caused on average an estimated 6 deaths per operational rocket just during its production stages.

Other non-astronaut fatalities

48 fatalities.

See also

Criticism of the Space Shuttle program

Effect of spaceflight on the human body

Fallen Astronaut

International Association for the Advancement of Space Safety

Laika

Maintenance of the International Space Station

Skylab 4

Notes

References

External links

The Encyclopedia Astronautica

Manned space programs accident/incident summaries (1963–1969) – NASA report (PDF format)

The Crash Site of the X-15A-3

Manned space programs accident/incident summaries (1970–1971) – NASA report (PDF format)

Interactive Space Shuttle Disaster Memorial

Raw Video Reconstruction of Space Shuttle Columbia Re-entry and More

Significant Incidents & Close Calls in Human Spaceflight